

# SLEEPER MODULE – SCOPE VS. CHANGE ANALYSIS

Change Analysis – Points 1 to 12

| Sr. No. | Module / Feature                     | Change Requested                                               | Scope Status            | Scope Basis                                                                                                                                          | Required New Development                                                                                                                        |
|---------|--------------------------------------|----------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | PO Assigned to Vendor                | Double Case Number Functionality                               | Not Explicitly in Scope | No specific double/split case-number requirement is stated.                                                                                          | NEW DEVELOPMENT – Double Case Number workflow and business logic.                                                                               |
| 2       | PO Assigned to Vendor / Call Raising | Multi-Batch Sleeper Call Raising Logic                         | Partially Covered       | Scope requires lot-wise, PO-wise and call-wise structures, but does not state multi-batch grouping in one inspection call.                           | NEW DEVELOPMENT – Multi-batch Sleeper Call Raising logic.                                                                                       |
| 3       | Inventory Management System          | New Cards: RM Used & RM Inventory Register                     | Partially Covered       | Raw Material Inspection is in scope, but dedicated RM Used and RM Inventory Register/consumption logic is not specified.                             | NEW DEVELOPMENT – RM consumption and inventory register cards.                                                                                  |
| 4       | Production Declaration               | Sleeper Category Classification (Mainline / Turnout / Special) | Not Explicitly in Scope | No sleeper-category-specific Production Declaration classification or category-based validation is specified.                                        | NEW DEVELOPMENT – Sleeper Category Classification.                                                                                              |
| 5       | Production Declaration               | Mould Sequence Management                                      | Not Explicitly in Scope | No mould sequence/bed-level lineage requirement is specified.                                                                                        | NEW DEVELOPMENT – Mould Sequence Management.                                                                                                    |
| 6       | Production Declaration               | PO Addition in Production Declaration                          | Partially Covered       | PO-wise data and automatic PO/inspection-call mapping are in scope, but PO selection during Production Declaration is not stated.                    | NEW DEVELOPMENT – PO Addition in Production Declaration.                                                                                        |
| 7       | Production Declaration               | Multiple Benches in Same Steam Curing Chamber                  | Not Explicitly in Scope | No many-to-one casting bench to steam chamber mapping or combined cycle monitoring is specified.                                                     | NEW DEVELOPMENT – Multiple Benches in Steam Curing Chamber.                                                                                     |
| 8       | Shift Duty Management                | Shift Duty Workflow & Logic Revamp                             | Partially Covered       | Shift-based duty allocation is explicitly in scope, but the requested dynamic roster, handover, attendance and supervision workflow is not detailed. | NEW DEVELOPMENT – Shift Duty Workflow & Logic Revamp.                                                                                           |
| 9       | Production Verification              | Unlock for Modification Workflow                               | Partially Covered       | Audit logs for update actions are in scope, but an authorized unlock-and-correct workflow with mandatory remarks is not specified.                   | NEW DEVELOPMENT – Unlock for Modification workflow.                                                                                             |
| 10      | List of Calls Pending                | Integrated Document Downloads (PO, MA, TC, Call Letter)        | Not Explicitly in Scope | Reports/documents are in scope generally, but direct one-click PO/MA/TC/Call Letter downloads from the pending-call list are not specified.          | NEW DEVELOPMENT – Integrated Document Downloads.                                                                                                |
| 11      | RM Inventory Management              | IE Raw Material Verification & Audit Flow                      | Partially Covered       | Raw Material Inspection and audit logging are broadly covered, but synchronized inward-vs-consumption-vs-balance verification is not specified.      | NEW DEVELOPMENT – IE RM Inventory Verification & Audit Flow.                                                                                    |
| 12      | Final Inspection - Card 1            | Visual Check & Measurement Workflow & Validation               | Partially Covered       | Final Inspection is in scope, but the detailed IRS T-39 visual/measurement validation rules are not specified.                                       | NEW DEVELOPMENT – Visual Check & Measurement Validation. Also includes multiple validation and workflow changes for this Final Inspection card. |

# SLEEPER MODULE – SCOPE VS. CHANGE ANALYSIS

Change Analysis – Points 13 to 23

| Sr. No. | Module / Feature              | Change Requested                                                              | Scope Status            | Scope Basis                                                                                                                                                                                                                             | Required New Development                                                                                                                             |
|---------|-------------------------------|-------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13      | Final Inspection - Card 2     | Epoxy Treated Sleepers Verification                                           | Not Explicitly in Scope | No epoxy-treated sleeper workflow, DFT, resin/hardener batch validation or serial mapping is specified.                                                                                                                                 | NEW DEVELOPMENT – Epoxy Treated Sleeper Verification. Also includes multiple validation and workflow changes for this Final Inspection card.         |
| 14      | Final Inspection - Card 3     | Critical Dimension Testing & Tolerance Engine                                 | Partially Covered       | Final inspection and mandatory validations are in scope, but the listed dimension-specific tolerance engine is not detailed.                                                                                                            | NEW DEVELOPMENT – Critical Dimension Tolerance Engine. Also includes multiple validation and workflow changes for this Final Inspection card.        |
| 15      | Final Inspection - Card 4     | Non-Critical Dimension Testing Engine                                         | Partially Covered       | Final inspection and validations are in scope, but section-wise non-critical dimension checks are not detailed.                                                                                                                         | NEW DEVELOPMENT – Non-Critical Dimension Testing Engine. Also includes multiple validation and workflow changes for this Final Inspection card.      |
| 16      | Final Inspection - Card 5     | Water Cube Strength Testing & 28-Day Curing Rules                             | Partially Covered       | Final inspection and validation are in scope, but 7/14/28-day cube rules and automated qualification thresholds are not specified.                                                                                                      | NEW DEVELOPMENT – Water Cube Strength Testing & 28-Day Rules. Also includes multiple validation and workflow changes for this Final Inspection card. |
| 17      | Final Inspection - Card 6     | Moment of Resistance (MOR) Testing & Verification                             | Partially Covered       | Final Inspection/testing is in scope, but MOR positions, calculations, IRS T-39 thresholds and re-test workflow are not specified.                                                                                                      | NEW DEVELOPMENT – MOR Testing & Verification. Also includes multiple validation and workflow changes for this Final Inspection card.                 |
| 18      | Final Inspection - Card 7     | Modulus of Failure (MOF) Testing Validation                                   | Partially Covered       | Final Inspection/testing is in scope, but structured MOF load-deflection and failure classification are not specified.                                                                                                                  | NEW DEVELOPMENT – MOF Testing Validation. Also includes multiple validation and workflow changes for this Final Inspection card.                     |
| 19      | Final Inspection - Card 8     | Modulus of Rupture (MOR/Rupture) Testing Logic                                | Partially Covered       | Final Inspection/testing is in scope, but rupture calculation and third-point loading validation are not specified.                                                                                                                     | NEW DEVELOPMENT – Modulus of Rupture Testing Logic. Also includes multiple validation and workflow changes for this Final Inspection card.           |
| 20      | Demoulding Inspection         | Demoulding Inspection Logic & Defect Rejection Rules                          | Partially Covered       | Process Inspection is in scope, but the specific demoulding defect breakdown and automatic batch yield calculation are not specified.                                                                                                   | NEW DEVELOPMENT – Demoulding Inspection Logic & Defect Rejection Rules.                                                                              |
| 21      | Annexures & Other Documents   | Document Upload Functionality – Frontend Service, Upload API & Database Table | Not Explicitly in Scope | The Scope mentions reports, documentation and API development, but does not specifically define an annexure/other-document upload service, dedicated upload API and supporting database table.                                          | NEW DEVELOPMENT – Frontend upload service, new document-upload API and database table for annexures/other documents.                                 |
| 22      | Closed Calls Card             | Closed Calls Card to Display Closed Call List                                 | Partially Covered       | The Scope includes inspection status dashboards, drill-down reporting and inspection workflows, but a dedicated Closed Calls card/list is not specifically defined.                                                                     | NEW DEVELOPMENT – New frontend Closed Calls card and service/API support to display the closed call list.                                            |
| 23      | Send to IBS / IBS Integration | Send to IBS Transaction & Status Synchronization                              | Partially Covered       | IBS integration, secure REST APIs and export of inspection results are in scope, but the specific Send to IBS transaction, new status workflow, backend/IBS status changes and supporting database tables are not specifically defined. | NEW DEVELOPMENT – Send to IBS transaction, frontend/backend service changes, new status handling, IBS changes and supporting database tables.        |